Guide to Writing Reports in LaTeX

by David Walk

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Preface

This document is a starters guide for LaTeX and should help you write your first reports. The goal is to get you over the first hurdles when learning LaTeX and help you focus on writing the report rather than fighting the editing software. This guide will be rather bare bones so you can focus on the relevant stuff. In some cases it might only guide you in the right direction and require you to get help from somewhere else to fix your problem. There are also several tips and tricks mentioned, which help with the general workflow of scientific writing, so this is not solely a LaTeX tutorial. In my opinion, LaTeX is best learnt by using a good template and changing/expanding it to ones needs over time.

It is assumed you are using the `reportDW` template or have added the same packages. There are many great LaTeX editors and local editors definitely have their advantages, but `Overleaf` will be used for this guide as it is widely used and easily allows collaborating with other students.

I also want to add some words of encouragement. When I started my studies, I didn't even know what LaTeX is. And here I am writing a guide for it a few years later. LaTeX can be a bit frustrating sometimes, especially for people who don't enjoy coding. But I think it is worth the effort as it allows creating beautiful documents. It is also possible with programs like Word, but you will see throughout this document all the quality of life features that come with LaTeX. If you manage to write one report with LaTeX, you will probably never look back.

`reportDW.cls`

Custom commands added by the template are marked in a blue box such as this one

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1 Getting Started

1.1 Upload Template

Upload the template (`.zip` file) to Overleaf via `New Project` → `Upload Project`. It is probably best to keep the first upload as the template and copy that project for every report.

1.2 Overview of the Files

Once inside your project, you can display the file tree on the left. You should see 3 files: `literature.bib`, `main.tex` and `reportDW.cls`. A quick summary of what they do:

`literature.bib` Contains all your bibliography entries. You add books, papers, and websites here, and LaTeX will automatically format the reference list for you. For most of the papers you can download the `bibtex` entry directly from the journal website.

`main.tex` The main document file. This is where you write the actual content of your report.

`reportDW.cls` The class file. You do not need to edit this; it defines the visual style and layout of the document behind the scenes. This is where you make changes if you want to make changes to the layout of the document.

1.3 Title Page

All relevant information for the title page is taken from the commands above `\begin{document}`. For example, you need to write your abstract inside `\abstract{write your abstract here}`.

Remark

Everything before `\begin{document}` is called the preamble.

2 Main Part

With the formatting all done, you can now focus on your report. This section will guide you through things like adding equations, figures, references, citations, etc.

2.1 Basic Text Formatting

Basic text formatting in LaTeX is straightforward. The most common commands are `\textbf{bold}` for **bold** and `\textit{italic}` for *italic* text. New paragraphs are started by leaving a blank line in the source.

Sections and subsections are added with `\section{}`, `\subsection{}` and `\subsubsection{}`. LaTeX numbers and formats them automatically.

Tip

For anything beyond this, a quick search will almost always give you the answer immediately. LaTeX is extremely well documented online.

Tip

A regular space in LaTeX can fall on a line break, which looks bad between a number and its unit or before a citation number. Use a tilde ~ instead to produce a non-breaking space. Use them for spacing units and citations: 9.81~m/s^2 and as shown by Smith~`\cite{smith2020}`.

2.2 Equations

Equations are one of LaTeX's greatest strengths. For inline equations, wrap your expression in dollar signs: `$E = mc^2$` renders as $E = mc^2$. For displayed, numbered equations use the `equation` environment

```
\begin{equation}
    E = mc^2
    \label{eq:einstein}
\end{equation}
```

This results in

$$E = mc^2 \tag{1}$$

Inside math mode, everything is italic. To add non-italic text, you have to wrap it inside `\text{}`.

`reportDW.cls`

The `mhchem` package is included in the template and provides the `\ce{}` command for chemical formulas. For example, `\ce{H2SO4}` produces H_2SO_4 and `\ce{CO2 + H2O -> H2CO3}` produces $CO_2 + H_2O \longrightarrow H_2CO_3$, a properly formatted reaction equation. Subscripts, superscripts, and arrows are all handled automatically.

2.3 Figures

Figures are included with the `figure` environment from the `graphicx` package. Upload your images to the project and ideally place them inside the `Figures` directory. You can then add those images to your document using

```
\begin{figure}[H]
    \centering
    \includegraphics[width=0.8\linewidth]{Figures/myimage.pdf}
    \caption{A descriptive caption goes here.}
    \label{fig:myimage}
\end{figure}
```

Tip

Whenever possible, use vector-based graphics (PDF, EPS, SVG, etc.) rather than raster formats like PNG or JPEG. Vector graphics scale without loss of quality and produce sharper output in print.

The placement of figures (and tables) can be challenging. By default (so without the `[H]`), LaTeX places your figures wherever it thinks they should go. The `[H]` forces the figure to be at the exact position where you add the `figure` environment.

2.3.1 Subfigures

In scientific papers, multi-panel figures conventionally have a bold letter label in the top-left corner of each panel. One way to achieve that without adding the letters to each figure by hand is using the subfigure package. You can move the `\caption` above the `\includegraphics` and leave the caption empty. This adds a letter above the subfigure.

```
\begin{figure}[H]
    \centering
    \begin{subfigure}{0.45\linewidth}
        \caption{}
        \includegraphics[width=\linewidth]{Figures/panel1.pdf}
        \label{fig:panel1}
    \end{subfigure}
    \begin{subfigure}{0.45\linewidth}
        \caption{}
        \includegraphics[width=\linewidth]{Figures/panel2.pdf}
        \label{fig:panel2}
    \end{subfigure}
    \caption{Overview of results.}
    \label{fig:overview}
\end{figure}
```

You can reference the subfigures in the text using their label. However, the letter is placed above the image rather than in the top-left corner, which does not match the convention used in most scientific journals.

reportDW.cls

The template alternatively provides `\labeledgraphic{width}{filename}{label}`, which overlays the letter directly onto the top-left corner of the image using `TikZ`. This matches the style commonly seen in journal figures. However, it requires sufficient white space in the top-left corner of the image. For figures where that area is occupied, the `subfigure` approach above is the safer choice.

```
\begin{figure}[H]
    \centering
    \labeledgraphic{0.45\linewidth}{Figures/panel1.pdf}{fig:panel1}
    \labeledgraphic{0.45\linewidth}{Figures/panel2.pdf}{fig:panel2}
    \caption{Overview of results. (a) First result. (b) Second result.}
    \label{fig:overview}
\end{figure}
```

Each panel is automatically labeled with a letter (a), (b), etc. in the top-left corner using the `TikZ` package and can be referenced individually with `\Cref{fig:panel1}`, which produces “Fig. 1a“. But there is also a downside to this approach. The letters are placed inside the image, which works well for plots with enough white space on the top-left but this approach fails in other cases.

2.4 Tables

Tables are created with the `table` and `tabular` environments together. Tables are a bit painful in LaTeX. I suggest using online tools and table converters to create the main part of the table and then change the `tabular` settings to get the desired formatting. Tables need some time to get used to and are probably the hardest part of learning LaTeX.

```
\begin{table}[H]
    \centering
    \caption{Experimental conditions for each sample.}
    \begin{tabular}{lcc}
        \toprule
        Sample & Temperature (\si{\celsius}) & Pressure (\si{\bar}) \\
        \midrule
        A & 25 & 1.01 \\
        B & 100 & 2.50 \\
        \bottomrule
    \end{tabular}
    \label{tab:conditions}
\end{table}
```

The letters in `{lcc}` define the column alignment: `l` for left, `c` for centre, and `r` for right. Again, the `[H]` is added to force the table to be rendered at that exact position in the PDF.

Tip

You can move the caption of tables above the table by moving the `\caption{}` above the `tabular` environment. Table captions should be above the table in scientific work.

`reportDW.cls`

Several column types are added by the template, namely `C`, `L` and `R`. These allow you to align the column as you want but also define the column width via e.g. `C{0.2\linewidth}`.

2.5 References

Any numbered element — equations, figures, tables, sections — can be cross-referenced. First attach a label with `\label{key}`immediately after the element, then refer to it with `\Cref{key}`. The `cleveref` package automatically prepends the correct word, so `\Cref{eq:einstein}` produces Eq. 1 (the template removes the parentheses around the number for equation references). Combined with the `hyperef` package, the whole Eq. 1 becomes a clickable hyperlink. In default LaTeX, one would use `\ref{eq:einstein}` or `\eqref{key}` for equations, which only generate the number. LaTeX updates all numbers automatically, so you never need to renumber manually.

Tip

It is good practice to use a consistent prefix convention for your labels, e.g. `eq:`, `fig:`, `tab:`, `sec:`, so they are easy to find in the source. Look at the examples above for ideas.

`reportDW.cls`

Two additional referencing commands are defined for convenience: `\pcref{key}` produces (Eq. 1) and `\lilyref{key}` produces (see Eq. 1), useful for parenthetical references mid-sentence such as "Einstein showed that mass and energy can be converted into one another (see Eq. 1)".

2.6 Citations

Citations are managed through the `literature.bib` file. You have to add all the sources you want to cite here as a `BibTeX` entry. An example entry looks like this:

```
@Book{prakBuch,
    author = {Erich Meister},
    title = {Praktikum Physikalische Chemie},
    publisher = {vdf Hochschulverlag AG an der ETH Zürich},
```

```
    year = {2022},
    chapter = {23},
}
```

In this example, `prakBuch` is the reference key. To cite it in the text, write `\cite{prakBuch}`, which produces a bracketed reference number and adds the source to the bibliography automatically. For most papers you can download the BibTeX entry directly from the journal website. Just find a `Export Citation` option and then choose BibTeX. In the rare case BibTeX is not available, choose for example ris and use an online converter to get the BibTeX entry.